

Week 1

```
GNU nano 8.7.1 week-1.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    char command[50];
    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
        exit(1);
    } else if (pid == 0) {
        // Child process
        printf("Child PID: %d, Parent PID: %d\n", getpid(), getppid());
        execlp(command, command, NULL);
        printf("Exec failed\n"); // runs only if exec fails
        exit(1);
    } else {
        // Parent process
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child completed\n");
    }

    return 0;
}
```

```
adith@KesavaSatish:~$ gcc week-1.c
adith@KesavaSatish:~$ ./a.out
Enter a Linux command: ls
Parent PID: 714
Child PID: 715, Parent PID: 714
a.out      arithmetic.c  p1.c.save  p3.c  p4.c.save  p6.c  p8.c      process-3.c  switch.c  week-1.c  week-4.c
arithmetic  p1.c          p2.c      p4.c  p5.c      p7.c  process-2.c  switch      three.c   week-3.c  zombie.c
Child completed
adith@KesavaSatish:~$
```